

	on:	
4(a)(i)	The magnetic force provides the centripetal force required to keep the particle in a circular path. $Bqv = \frac{mv^2}{r}$ $r = \frac{mv}{Bq}$ $= \frac{(9.11 \times 10^{-31})(5.0 \times 10^7)}{(1.2 \times 10^{-3})(1.60 \times 10^{-19})}$ $= 0.2372$ $= 0.24 \text{ m}$	[1] for correct expression and numerical substitution [1] for correct answer

No.	Solution	Remarks
4(a)(ii)	No. The plane of the circular path is perpendicular to the axis of the solenoid. <u>Common mistakes:</u> - 0.24m is larger than 0.20m, so will collide	[1] for correct answer [1] for correct explanation
4(b)	As the radius decreases, the <u>speed decreases</u> too. The electron could be <u>losing kinetic energy</u> due to collision with air molecules (i.e. the space is not a perfect vacuum). <u>Common mistakes:</u> - Air resistance caused the speed to decrease (air resistance is a resistive force that acts on a solid that is much larger). - The path is helical.	[1] for correct answer [1] for correct reason
4(c)	$F_B = F_E$ $Bqv = qE$ $E = Bv$ $= (1.2 \times 10^{-3})(5.0 \times 10^7)$ $= 6.0 \times 10^4 \text{ N C}^{-1}$	[1] for correct expression and numerical substitution [1] for correct answer
5(a)(i)	Work function energy is the minimum amount of energy needed to	[1] for correct